

Automated Topology Generation for Power Amplifier Designs through BiLSTM-based DNN and Multi-objective Optimizations

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Abstract—This work presents an automated, intelligent methodology for optimizing power amplifier (PA) design by predicting the most suitable circuit topology—specifically, the input and output matching networks—for a given high electron mobility transistor (HEMT). A classification-based bidirectional long short-term memory (BiLSTM) deep neural network (DNN) is trained to determine the optimal PA topology, while multi-objective Pareto front-based optimization techniques refine the network's hyperparameters, including the number of hidden layers and neurons. The proposed approach is adaptable to various HEMT models and is validated through the design and optimization of high-performance PAs using lumped elements and transmission lines, operating within the 1–2 GHz frequency range. The method is demonstrated using the Cree CGH40010 GaN HEMT on a Rogers RO4350B substrate, achieving a power output of approximately 40 dBm, a power-added efficiency (PAE) of at least 50%, and a power gain exceeding 10dB.

I. INTRODUCTION

The power amplifiers (PAs) are important circuits in the fifth-generation infrastructure, and for these nonlinear microwave designs, computer-aided design techniques play crucial roles in decreasing the designing time along with design cost [1]. In these PAs, various technologies for the active devices (i.e., high electron mobility transistor (HEMT)) are used. The recently published works employ the artificial neural network (ANN) for various purposes. In [2], transistor nonlinear behavioral models are introduced based on the ANN method. The harmonic prediction with the help of ANN is executed in [3], which is based on the nonlinear characteristics and memory effect of PA. The long short-term memory (LSTM)-based ANN is used in [4] for modeling the transient response of a low-noise amplifier design. For the first time, we propose an automated methodology leading to predict the optimal structure of PA for the executed HEMT transistor through the classification of Bi-directional LSTM (BiLSTM)-based deep neural network (DNN) along with determining the optimal hyperparameters of trained DNN through multi-objective optimization methods. The proposed method is validated by firstly predicting the optimal matching networks (MNs) of a PA with lumped elements (LEs), and afterward, the transmission line (TL)-based PA is optimized in terms of power gain (G_p), output power (P_{out}), power added efficiency (PAE), and phase distortion which shows a remarkable agreement.

II. PROPOSED METHODOLOGY IN A NUTSHELL

Constructing the initial structure of PA for the executed HEMT device depends heavily on the experience of designers.

Hence, our proposed methodology is a flexible “Black-Box” modeling, which leads to predicting the optimal input and output MNs for the gallium nitride (GaN) HEMT device, achieving targeted outcomes for the PA. This automated optimization process includes two phases operated concurrently: i) training a classification BiLSTM-based DNN for predicting the best topology of PA that fits the best for the determined HEMT device, as well as ii) optimizing the hyperparameters of trained DNN through multi-objective optimizations results in an appropriate accuracy. This section is devoted to presenting the summary of the proposed automated approach, and the related flowchart is depicted in Fig. 1.

The automated nonlinear optimization method starts by connecting the electronic design automation (EDA) tool (here, Keysight ADS) and numerical analyzer (here, MATLAB), operating collaboratively and sequentially (Step-1). Afterward, the recommended data of selected GaN HEMT are extracted from load pull simulation in ADS, and the HEMT datasheet is used as input data for the trained DNN (Step-2). Then, the simplified real frequency technique (SRFT) is employed, leading to generating input and output MNs varied in the number of components [5]. By employing the SRFT, n and m , different types of input and output MNs are generated, respectively (Step-3). Hence, in total, $n \times m$ different PA structures are generated, and the question emerges: What is the best-fitted PA structure for the executed HEMT device? For this case, the proposed BiLSTM-based DNN can be useful enough to predict the most suitable topology of PA among $n \times m$ structures. In Step 4, a classification BiLSTM-based DNN is constructed for which the softmax layer is employed as an activation function, and the cross entropy layer is executed as a loss function.

Along with training the DNN, determining the optimal hyperparameters for the constructed network is also significant. For this purpose, multi-objective optimizations based on the Pareto optimal front (POF) as multi-objective Pareto front using modified quicksort (PFUMQ), Thompson sampling efficient multiobjective optimization (TSEMO), and multi-objective ant lion optimizer (MOALO) algorithms [6] are employed (Step-5).

III. PRACTICAL IMPLEMENTATION OF PROPOSED METHOD

The proposed automated method is started and validated by selecting ‘Cree CGH40010’ as the GaN HEMT device, and the substrate used is Rogers RO4350B. The SRFT method is employed and results in 28 different LE-based PA structures.

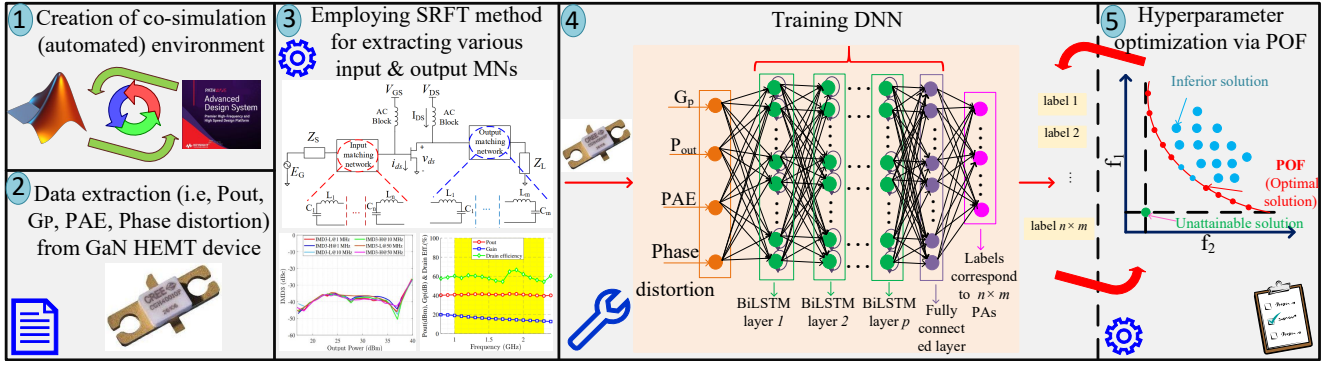


Fig. 1. Proposed automated methodology for predicting the optimal PA structure for the determined GaN HEMT device.

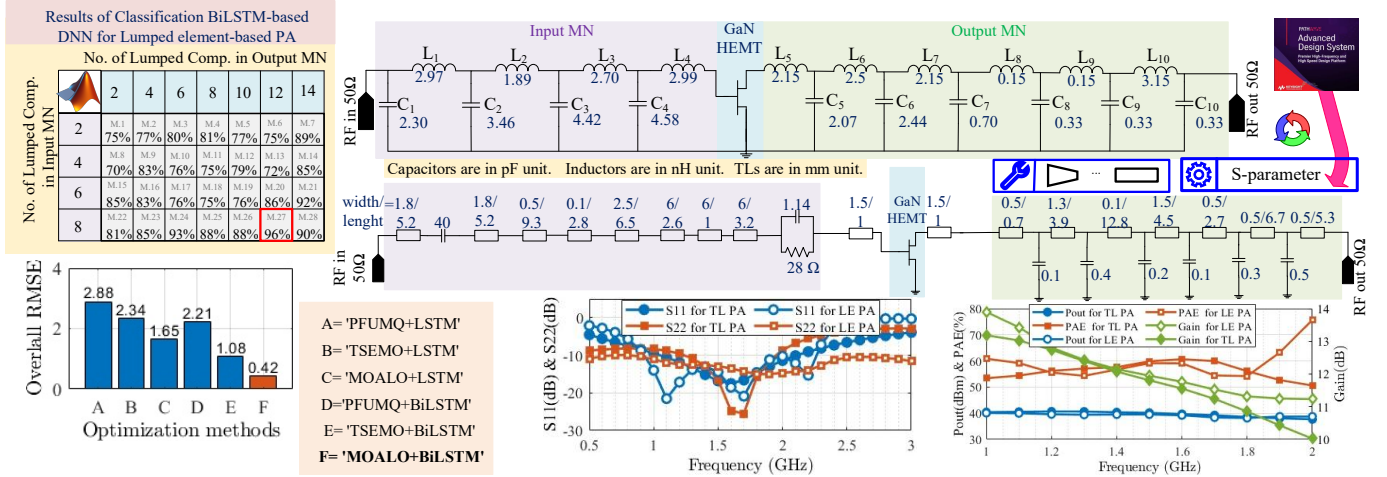


Fig. 2. Practical implementation of an automated method for LE- and TL-based PAs used GaN HEMT with the simulation results.

A suitable amount of training data is obtained by iterating randomly the initial design parameters achieved from the SRFT method, and in total, 2000 multi-segment data are collected. As Fig. 2 shows for the inserted transistor, the BiLSTM-based DNN is trained, and 27th model ($n=8$ and $m=12$) has the 96% most fitted performance. The hyperparameter optimization is executed for the nominated POF functions with both LSTM and BiLSTM networks, which shows that the MOALO function with the use of BiLSTM layers achieves the perfect result in comparison with others; the trained DNN includes 4 hidden layers with 250 neurons in each one. By the selected LE-based PA, the 'S-parameter' simulation is executed in ADS leads to be matched to the TL components. As Fig. 2 depicts, LE-based PA and TL-based PA meet perfectly with each other operating from 1 GHz to 2 GHz, at P_{out} of around 40 dBm, with minimum PAE of 50%, and a G_p larger than 10 dB.

IV. CONCLUSIONS

This paper presents an automated methodology leveraging a BiLSTM-based deep neural network (DNN) to predict the optimal power amplifier (PA) topology for a given GaN HEMT device. In addition to training the DNN, hyperparameter optimization is performed using multi-objective optimization techniques to enhance model accuracy and performance. The proposed approach is validated through the design and optimization of high-performance PAs operating within a 1 GHz bandwidth. Re-

sults demonstrate that the MOALO method, combined with the BiLSTM architecture, effectively optimizes PA design, making it a promising solution for high-dimensional PA architectures.

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